Tushar Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.66
Delhi Public School Ranipur	Class XII (CBSE)	2023	95.00%
Delhi Public School Ranipur	Class X (CBSE)	2021	95.60%

EXPERIENCE

Software Engineering Intern | Kotak Mahindra Bank

Microservice creation for customer service:

- Designed and deployed a Spring Boot microservice from scratch handling WhatsApp consent workflows for 6000+ customers daily.
- Created and configured AWS secrets for the service and added update pipeline for CI/CD.
- Built Grafana dashboards monitoring latency, error rates and service health, reducing debugging time by 50% by implementing email alerts on cascading failures.

PDF Generation using Playwright:

- Developed PDF generation service using Playwright and Chromium, eliminating commercial licensing dependency and reducing annual software costs by 50lakh.
- Used java design patterns(factory and template), and SOLID principles for the design. Used docker image for CI/CD and security.

PROJECTS

Financial AI Coach | Java ,Springboot, PostgreSQL, React

- Built AI-powered financial copilot processing 5k+ mock transactions and generating personalized financial insights using LLM workflows.
- Implemented unit and integration tests using JUnit, achieving 85% code coverage through Test-Driven Development.

ML Algorithm Library | Python, NumPy, Pandas, Scikit-Learn

- Implemented AdaBoost, Random Forest, SVM, Decision Trees and Linear Regression from scratch without ML libraries, achieving within 90% accuracy of Scikit-Learn implementations across benchmark datasets.

Emotion Detection Using Face Recognition | Python, NumPy, OpenCV, Tensorflow

- Built a custom pipeline by integrating a face detection model with a trained emotion classification model. Achieved real-time emotion recognition, enhancing model performance through data preprocessing and tuning.

Linear Algebra & Differential Equation Solver App | Course Project | Python, NumPy, Pandas, CustomTkinter

- Implemented algorithms like LU and QR decomposition, orthogonal collocation, and finite element methods to solve linear algebra and differential equations. Built an interactive GUI for efficient input handling and solution visualization, streamlining complex numerical computations.

POSITIONS OF RESPONSIBILITY

Placement Coordinator | Office of Career Services, IIT Hyderabad

Internship Coordinator | Office of Career Services, IIT Hyderabad

Sponsorship Coordinator | SPIC MACAY, IIT Hyderabad Hobbies and Interests

SKILLS

Programming Languages: Java, C/C++, python, SQL

Cloud and DevOps: Docker, Kubernetes, AWS, Azure DevOps

Backend: Spring Boot, REST APIs, Microservices

Data & ML: TensorFlow, Scikit-Learn

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOPs

Soft Skills: Effective Communication, Team Collaboration, Problem Solving, Adaptability, Time Management

RELEVANT COURSES

Mathematics: Calculus-I & II, Differential Equations, Elementary Linear Algebra, Numerical Methods (Basic and Advanced), Introduction to Stochastic Differential Equations, Applied Mathematics in Chemical Engineering

Soft Skills: Communication Skills, Communication Skills: Advanced

Programming: Introduction to Programming

HOBBIES AND INTERESTS

- I play variety of sports ranging from Volleyball and cricket to indoor games and video games.
- Reading is one of my hobbies, i like to read novels in Hindi and English language of different genre, sci-fi, non-fiction, fiction. I also read manga with my favorite being Bleach.
- Music is also one of my passions, hence i play a little bit of guitar.